# Project Proposal

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## Dataset Protoytpe

For this project, we plan to use historical daily adjusted closing prices directly from the Yahoo Finance API, covering the period from January 1, 1998, to the present. This timeframe was selected to encompass three major structural market shocks: the Dot-com Bubble, the 2008 Financial Crisis, and the 2020 COVID-19 crash. Our primary analysis will focus on the S&P 500 Index. To validate the accuracy of our approach, we may utilize the CBOE Volatility Index (VIX), allowing us to compare our model's estimated latent volatility against Wall Street's implied volatility. Additionally, assets such as sector-specific ETFs (XLK, XLF) and a safe-haven treasury bond (TLT) will be incorporated to test if the model's regime-switching logic generalizes reliably across different market segments, as well as sector-specific insight of each crisis. To reduce the computational burden while preserving essential volatility clustering, the daily data will be mutated to only include the final trading day of each week to establish consistent weekly boundaries. Finally, these weekly prices will be transformed into weekly log returns, calculated as $Y_t = \ln(P_t/P_{t-1})$. This transformation is critical, as it ensures astationary distribution of the time series which facilitates its modelling

### Which ticker is each index?

- ^GSPC (S&P 500 Index): Top 500 stocks in the U.S. This is the observation that we will most likely use.
- ^VIX (CBOE Volatility Index): Market volatility measure that we can use for model validation.
- XLK (Technology Select Sector SPDR Fund): An ETF containing all the tech stocks in the S&P 500. Maybe use for dot com crash?
- XLF (Financial Select Sector SPDR Fund): An ETF containing all the financial stocks in the S&P 500 (banks, insurance, etc.). Check for 2008 Financial Crisis?
- TLT (iShares 20+ Year Treasury Bond ETF): This tracks long-term US government bonds. "Safe haven" asset that normally behaves opposite to stocks. Does the model still work with this?

We may also choose to include some other indexes.

```r
project_tickers <- c("^GSPC", "^VIX", "XLK", "XLF", "TLT")
market_data <- tq_get(project_tickers,
                      get  = "stock.prices",
                      from = "1998-01-01")
```

```
## Warning: There were 5 warnings in `dplyr::mutate()`.
## The first warning was:
## i In argument: `data.. = purrr::map(...)`.
```

```
## Caused by warning in `zoo()`:
## ! some methods for "zoo" objects do not work if the index entries in 'order.by' are not unique
## i Run `dplyr::last_dplyr_warnings()` to see the 4 remaining warnings.
```

```r
weekly_returns <- market_data %>%
  mutate(week = floor_date(date, "week")) %>%
  group_by(symbol, week) %>%
  arrange(date) %>%
  slice_tail(n = 1) %>%
  ungroup()


weekyly_log_returns <- weekly_returns %>%
  group_by(symbol) %>%
  arrange(date) %>%
  mutate(
    previous_price = lag(adjusted),
    log_return = log(adjusted / previous_price)
  ) %>%

  filter(!is.na(log_return)) %>%
  select(symbol, date, adjusted, log_return)

head(weekyly_log_returns)
```

```
## # A tibble: 6 x 4
## # Groups:   symbol [2]
##   symbol date       adjusted log_return
##   <chr>  <date>        <dbl>      <dbl>
## 1 ^GSPC  1998-01-09    928.    -0.0498
## 2 ^VIX   1998-01-09     28.7    0.203
## 3 ^GSPC  1998-01-16    962.     0.0358
## 4 ^VIX   1998-01-16     21.6   -0.282
## 5 ^GSPC  1998-01-23    958.    -0.00409
## 6 ^VIX   1998-01-23     23.3    0.0743
```

## Scientific Questions

Which assumption—discrete regime shifts (HMM) or continuous volatility evolution (Stochastic Volatility)-better explains the market dynamics during the Dot-com bubble, the 2008 Financial Crisis and the COVID-19 crisis?

## Hidden Markov Model

### The Likelihood

Let $Y_t$ represent the weekly log return of the index at time $t$, and let $X_t \in \{1, 2\}$. State 1 represents a "Calm/Normal" market regime, and State 2 represents a "Crisis" regime.

$$Y_t \mid X_t = i \sim \mathcal{N}(\mu_i, \sigma_i^2)$$

Where:

$$\mu_1, \mu_2 \sim \mathcal{N}(0, 1)$$

$$\sigma_1, \sigma_2 \sim \text{Exponential}(1)$$

(Note: Should we enforce $\sigma_1 < \sigma_2$, ensuring State 1 always represents the lower-volatility normal market, and State 2 always represents the crisis?). 2. The Latent State Dynamics (The Regime Switching)The sequence of unobserved market states follows a first-order Markov process. The probability distribution of tomorrow's state, $X_t$, is explicitly conditioned on today's state, $X_{t-1}$.

**Latent States**

$$X_t \mid X_{t-1} = 1 \sim \text{Categorical}(P_{11}, 1 - P_{11})$$
$$X_t \mid X_{t-1} = 2 \sim \text{Categorical}(1 - P_{22}, P_{22})$$

**Transition Probabilities**

$$P = \begin{pmatrix} P_{11} & 1 - P_{11} \\ 1 - P_{22} & P_{22} \end{pmatrix}$$

where,

$$P_{11} \sim \text{Beta}(1, 1)$$
$$P_{22} \sim \text{Beta}(1, 1)$$

## Stochastic Volatitily Model

## Equal Contribution